

Problem Statement 2: Smart Commuter Companion

Comprehensive Problem Specification, Data Architecture & Judging Framework

Target: Mobile-first web app User: The commuter Personas: 3 (pick one) Scoring: 100% / 3 criteria

1. Challenge Statement

THE MANDATE

Build a smart commuter companion mobile app that delivers proactive decision support during planned and unplanned events, tailored to commuters' needs.

Four words in that sentence carry the whole brief, and each one is scored.

- **Proactive:** It reaches the commuter before the problem does, not after. A push at 07:30 about the 08:15 platform beats a banner at 08:16.
- **Decision support:** It recommends an action, not just a status. "Take the 190 from Bt Batok, +11 min" beats "NSL delays".
- **Planned and unplanned:** Scheduled works, early closures and maintenance windows are known in advance and are just as disruptive. Treating them as first-class is a differentiator, not an afterthought.
- **Tailored:** The same disruption means different things to Rachel, Arjun and Mdm Lim. Generic output serves nobody -- see Section 2.2.

2. Challenge Details

2.1 Introduction

Singapore's transport network works well on an ordinary day. The commuter's problem is the day that is not ordinary -- a signalling fault at 08:15, a station exit closed for works, a line running at reduced frequency, a downpour that turns a 6-minute walk into a decision.

Today the burden of reacting falls on the commuter: notice something is wrong, work out whether it affects them, decide what to do instead, and do it while standing on a platform. Most apps are reactive and generic -- they tell everyone the same thing, after the fact.

We want the opposite. An app that knows enough about this commuter's routine to tell them, before they leave, that today is different and here is what to do about it.

2.2 The Commuter

The user here is the commuter, not an operator. Three personas define what that means in practice. You do not need to serve all three well -- but say which one you are building for, and show a real journey working end to end for that person.

Persona	Journey	What They Need
Rachel -- fixed-schedule commuter	Tampines to Raffles Place, EWL, leaves 07:40, at her desk by 08:45	Same trip for four years; checks no app on a normal day. Five minutes is noise, fifteen costs her a meeting. Interrupt her only when it matters, in one line.
Arjun -- multi-modal, flexible start	Punggol to one-north; cycles to the LRT, sometimes buses the whole way	Start time flexible within about an hour. Optimises comfort over speed and will leave twenty minutes later to avoid a crush. Cares about crowding, shelter, and his bike.
Mdm Lim -- accessibility-constrained	Bedok to Singapore General Hospital, fortnightly appointment	Walks slowly, avoids stairs, needs lifts and sheltered walkways, will not improvise on the platform. Needs large text, both walking legs planned, and a day-before lift/exit warning.

You may propose your own persona if you can justify it, but do not silently build for "a generic commuter" -- that is how apps end up serving nobody.

2.3 The Geospatial Base (OpenStreetMap)

OpenStreetMap is the required geospatial base -- see Section 3.2.2 for the requirement itself. It is open, it is licensed for this use, and it carries the pedestrian and cycling detail this problem needs -- footways, crossings, stairs, lifts, covered walkways, cycle paths -- that a plain road map does not.

Need	Option
Bulk Singapore data	Geofabrik extract -- download.geofabrik.de/asia/malaysia-singapore-brunei.html
Targeted queries	Overpass API -- overpass-api.de (build queries interactively at overpass-turbo.eu)
Routing engines	OSRM (project-osrm.org), GraphHopper, or Valhalla -- all open source, all self-hostable, all consume OSM data
Map rendering	Leaflet (leafletjs.com) or MapLibre GL JS (maplibre.org)

TWO CONDITIONS ON OSM USE

Attribution is required -- OSM data is licensed under the ODbL, so display "(c) OpenStreetMap contributors" wherever you show the map or derived data. And do not hammer the public tile servers: tile.openstreetmap.org runs on donated infrastructure and its usage policy prohibits heavy or bulk use by applications. Use a free provider's own key, self-host tiles, or bundle a static extract. The same applies to the public Overpass instance -- cache what you fetch rather than querying it in a loop.

OSM is the required base, not an exclusive one. Layer LTA's DataMall geospatial layers on top of it where they are better -- CoveredLinkWay, TrainStationExit and CyclingPath are authoritative in a way crowd-sourced data is not. OneMap's routing API is also fair game alongside your own.

2.4 The Data

The participant brief (PS2_README.md, Section 2.4) carries the full data architecture: every relevant DataMall endpoint, the Train Service Alerts schema, the line-code inconsistency trap, the three separate crowding signals, and the data.gov.sg, OneMap and other government sources. This section summarises what is provided and where the rest comes from.

- **Provided in this repository:** PS2/data/AmendmenttoMP2014RailStation.geojson -- rail station geometry: 208 polygons. These are station footprints, not points; TYPE carries only MRT/LRT/CCL, not line codes; crs is null, so confirm the coordinate reference system before joining it to anything. PS2/data/UsefulWebsites.txt -- every external source named in this section, with its URL and what it is for. PS2/references/LTA_DataMall_API_User_Guide.pdf -- full DataMall API documentation, version 6.8. PS2/references/24hourWeatherForecast.json and 4dayWeatherForecast.json -- OpenAPI specifications for two data.gov.sg weather endpoints: interface documents, not weather data.
- **LTA DataMall:** Register for a free AccountKey at datamall.lta.gov.sg, then call datamall2.mytransport.sg/ltadatamall/<endpoint> with an AccountKey header. Responses page 500 rows at a time via \$skip.
- **The endpoint that matters most:** TrainServiceAlerts is the official structured train disruption feed and the single most important endpoint for this brief. Read its schema before you design anything -- note the per-disruption detail is nested inside AffectedSegments, and Message is a separate list carrying Content and CreatedDate.
- **A trap worth knowing early:** Line codes are not consistent across endpoints. Reconciling them is part of the work, not a bug in the data.
- **Crowding is three signals, not one:** PlatformCrowdDensityRealTime, PlatformCrowdDensityForecast and the bus-load field in BusArrival answer different questions. Do not treat them as interchangeable.

2.5 Ground Rules on Data Use

Other data is encouraged, with one condition: it must be lawfully obtainable and you must say where it came from.

- **Do not scrape in breach of terms.** A site's terms of service govern. Data obtained in breach of them caps the data dimension at 1 and is referred to the organisers before scoring continues.
- **Do not use personal data** you have not been given permission to use. If your app collects a user's routine or location, say in your submission what you store, where, and for how long. A commuter companion is a privacy-sensitive product by nature; treat it as one.
- **Do not commit credentials.** API keys go in environment variables or an ignored .env file, never in the repository.

2.6 Additional Notes

- **Underground, there is no signal.** A commuter between stations cannot fetch anything. Decide what your app does in that window -- cache the current journey, degrade gracefully, or say plainly that it is stale -- and state the choice in your write-up.

- **The feeds are quiet most days.** TrainServiceAlerts.AffectedSegments is empty on a normal day, so do not build a demo that depends on a real disruption occurring during judging. You may demonstrate the major-disruption path by replay or injected test data provided it is labelled as such. Everything else in the feed -- including its Message stream of live bus diversions and advisories -- is populated daily and needs no such allowance.
- **Ask if you are unsure.** Especially about a data source you are not certain is in bounds. We would rather answer a question than disqualify a good idea.

3. Expectations & Goals

3.1 Scope

BUILD A WEB APP, DESIGNED MOBILE-FIRST

No native build, no app store -- a web application a commuter opens in their phone's browser. Mobile-first means the phone is the primary target, not an afterthought you shrink down to. Judges will open your app in a mobile browser on a real phone, not in a resized desktop window, and score the experience there.

If it also works well on a desktop, good -- but a desktop layout squeezed onto a phone will not score. Design for the real conditions: a small screen, one thumb, a moving train, bright sunlight, an address bar that appears and disappears as you scroll, a soft keyboard that eats half the viewport, and -- underground -- no signal at all.

What we are not asking for is a dashboard for operators. The user here is the commuter.

3.2 Required Capabilities & Judging Rubric

Three things are mandatory. A submission missing any of them is incomplete, however good the rest is.

3.2.1 Route Planning

The app must plan an actual journey, not just report status. Given an origin, a destination and a time, it must produce a route the commuter can follow -- and when conditions change, it must produce a revised one.

- **Multi-modal** where the persona needs it: rail, bus, walk, and cycle for Arjun.
- **Door to door**, including the walking legs at each end. A route that starts at a station and ends at a station is not a commuter's journey.
- **Responsive to live conditions:** a disruption, a crowded platform or heavy rain should change what the app recommends, and the app should say why it changed.
- **Realistic timing**, with the uncertainty made visible rather than hidden behind a single confident number.

You may build routing on top of an existing engine -- see Section 2.3. We are not asking you to invent a shortest-path algorithm; we are asking for routing that reflects what is actually happening on the network today.

3.2.2 GIS on OpenStreetMap

Use OpenStreetMap as your geospatial base. Section 2.3 covers where to obtain it, the attribution requirement, and the tile-server usage policy. Both conditions in that section are requirements, not suggestions.

3.2.3 Visualisation

The commuter must be able to see the situation, not just read about it. The app needs a visual layer that makes the state of their journey legible at a glance on a phone screen.

- **The route on a map**, with the affected portion clearly distinguished from the unaffected portion.
- **The alternative shown against the original**, so the commuter can judge the trade-off rather than being told to trust the app.
- **Crowding readable in one second.** This is a three-level scale; it does not need a chart.
- **Time and delay shown** so the cost of each option is obvious.

TWO THINGS WE LOOK FOR SPECIFICALLY

It works on a phone, in motion, in one hand -- test it on an actual phone browser, because mobile emulation in devtools hides viewport-height, soft-keyboard and touch-target problems that only show up on the real thing. And every visual element earns its place -- a map showing dots with no context, or a chart nobody would act on, is worse than not having it: it costs screen space and attention that a commuter standing on a platform does not have.

Accessibility is part of this, not separate from it: legible type sizes, sufficient contrast, and not relying on colour alone to convey a state.

3.2.4 Judging Rubric

Submissions are graded against three criteria, weighted to 100%.

Criterion	Weight	What Judges Look For
Problem Fit	40%	Would a real commuter be better off with this app than without it? Persona fit, proactivity and the quality of the decision it offers -- plus whether the submission does anything beyond what this brief asked for.
Technical Execution	35%	Routing that reflects live conditions on an OpenStreetMap base, the breadth and judgement of the data brought in, and whether the thing is actually built -- runs from a clean README, performance is sane.
Ease of Use	25%	A commuter could pick it up on a phone, one-handed, and get an answer. Interaction design, information hierarchy and accessibility. Scored on a real phone browser, not desktop emulation.

Each criterion is scored on a 0-5 scale and weighted. Commuter value and design together carry more than half the total -- a technically impressive app that a commuter would not open twice cannot score well here, and that is deliberate.

- **Mandatory-capability cap.** A submission missing one of the three required capabilities in 3.2.1 to 3.2.3 cannot exceed level 3 on the part of the score that covers it.

Caps apply regardless of other merit. A feature shown in the pitch but absent from the running system, mocked data presented as live, a claim a judge cannot verify, a credential committed to the repository, or OpenStreetMap used without attribution -- each caps the part of the score it touches, and the attribution one is a licence breach rather than a style point.

How judging runs. Judges follow your README on a clean machine; the app is opened in a browser on a real phone; you walk one real journey for your chosen persona end to end; and you are asked to defend one claim per criterion. Saying it is in the slides is not evidence.

3.3 Beyond the Brief

Part of Problem Fit is whether a submission does anything beyond what this brief asked for. This section is where that is earned. It describes the problem as we currently understand it; a submission that surfaces a capability we did not ask for, and that on seeing it we would want, scores at the top of it.

None of it is required. A submission that answers the brief exactly and claims nothing beyond it can still score well.

3.3.1 Use of AI

USING AI IS NOT A REQUIREMENT -- BUT WHERE YOU USE IT, IT MUST EARN ITS PLACE

We expect most teams to use AI while building. That is fine and expected. What scores is AI that makes the product better for the commuter, not AI used for its own sake.

- **Structured advice from free text:** turning free-text service notices into structured, personalised advice.
- **Predicting duration or escalation:** how long a disruption will last, or the probability a delay escalates, so the app can advise "wait" versus "reroute".
- **Learning a routine:** inferring from travel history which disruptions actually matter to this individual.
- **Conversational or voice interaction** for hands-free use on a platform.
- **Ranking alternatives** against a persona's real preferences rather than shortest-time-wins.

An AI feature you can show works -- with a measurement, however simple -- is worth far more than one you can only describe. A well-argued decision not to use a model where a simpler method works is also creditable.

- **GCP credits are expected to be available** to participating teams; details will be confirmed separately. If you plan to depend on a specific GCP service, tell us early so we can make sure access is in place.
- **Reproducibility without payment:** a judge must be able to check it without paying for anything. If something can only be demonstrated or regenerated by spending money, it does not count.

3.3.2 Open Innovation

The brief describes the problem as we currently understand it. A submission that surfaces a capability we did not ask for -- and that, on seeing it, we would want -- scores at the top of that component.

To be creditable it must be sound enough to survive scrutiny: stated assumptions, known limits, and a plausible path to production. An original idea nobody can use is not yet a contribution -- Problem Fit is led by whether a real commuter is better off, and originality is judged in that light.

4. Deliverables

#	Deliverable	What It Must Contain
1	The app	Runnable, with clear setup instructions, covering all three required capabilities in Section 3.2. Judges will follow your README on a clean machine; what does not run does not score.
2	A short write-up	Which persona you built for, your architecture, your assumptions, and the limitations you know about. If you claim a number anywhere -- an accuracy figure, a speed-up, a comparison against an alternative -- say how you arrived at it. A claim a judge cannot check does not score.
3	A demo	Walk one real commuter journey, end to end, through one real disruption.

QUESTIONS

Ask the organisers -- especially about a data source you are unsure of. See Section 2.5, Ground Rules on Data Use. We would rather answer a question than disqualify a good idea.